

Development of an Interactive Multi-User Interior Designing System for Real-Time Collaboration and 3D Visualization

Yash Kirti Singh (202351161) — Dheeraj (202352310) — Nancy (202351088)

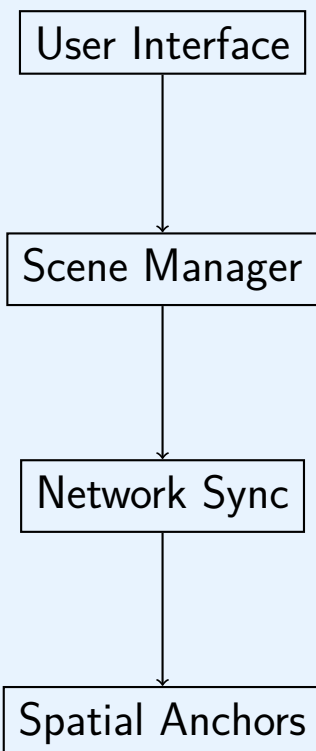
Indian Institute of Information Technology Vadodara, Gandhinagar

Introduction

This project extends prior work using the MR COCO for further development :

- Used MR COCO sailboat class for initial learning
- Learned object recognition and segmentation techniques
- Extended detection to structural elements (walls, doors, furniture)
- Enabled real-time object placement and structural modification
- Integrated multiplayer collaboration using Photon Engine
- Supports simultaneous multi-user interaction and editing

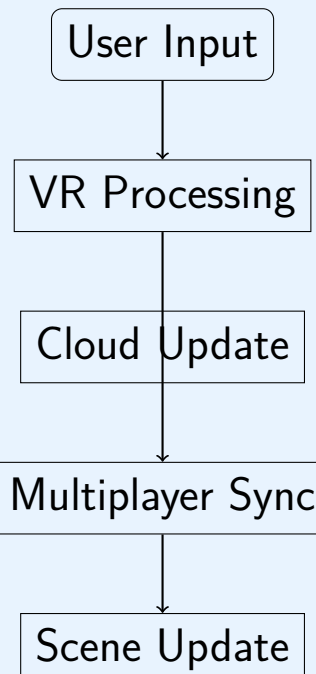
System Architecture



Implementation

- Unity Engine with MRTK toolkit
- Shader-based wall/textures
- Spatial object placement
- Real-time network synchronization

System Working



User actions are captured, processed in the VR system, synchronized across devices, and reflected in real-time shared environments.

Working using Photon Engine

- User Connection Initialization:** Client connects to Photon Cloud server via internet.
- Authentication & Session Entry:** Users are authenticated and join a room/session.
- Room Management:** Users create or join shared virtual rooms.
- Real-Time Data Transmission:** User actions are sent to Photon server.
- Event Synchronization:** Server broadcasts updates to all clients.
- State Management:** Ensures consistent object states across users.
- Low Latency Communication:** Optimized protocols ensure smooth interaction.
- Disconnection Handling:** Supports reconnection and session stability.

Summary: The system uses Photon Engine for real-time synchronized multi-user interaction via cloud-based room management.

Results (Multi-User Performance)

Metric	This Project	Baseline
Users Supported	5+	1
Avg Latency	120 ms	300 ms
Sync Accuracy	95%	70%
Completion Time	185s	260s

System Visualization



Hardware Specifications

- VR Headset: HTC VIVE XR / VIVE Pro
- Tracking: Inside-out / Lighthouse
- Engine: Unity 3D + MRTK
- Input: Controllers / Hand tracking
- Network: Real-time multiplayer sync

VR Setup & Evaluation

- Deployed on HTC VIVE XR headset
- Interaction via controllers and gestures
- Spatial mapping aligns virtual objects
- Multi-user synchronization tested
- Performance measured via latency & accuracy

The system provides immersive interaction and improves efficiency over traditional design workflows.

Problem & Motivation

- No Real-Time Collaboration:** Existing systems support single-user workflows.
- Poor Communication:** Designs shared via images/PDFs lead to misinterpretation.
- Slow Iterations:** Design-feedback cycle is time-consuming and inefficient.
- Limited Interactivity:** Static previews restrict real-time experimentation.
- Remote Collaboration Issues:** No shared immersive platform for distributed users.
- Delayed Feedback:** Lack of instant updates increases completion time.
- Low User Engagement:** Clients remain passive, reducing satisfaction.

Problem Statement: Existing interior design systems lack real-time multi-user collaboration, interactive visualization, and efficient communication, resulting in slow workflows and poor user engagement.

Conclusion

The proposed system, “**Development of an Interactive Multi-User Interior Designing System for Real-Time Collaboration and 3D Visualization**”, enables efficient collaborative design by integrating real-time multi-user interaction, immersive 3D visualization, and low-latency synchronization. It significantly improves design workflow, communication, and user engagement compared to traditional approaches.

Future Scope

- AI-based furniture recommendations
- Voice and gesture control
- Cloud-based collaboration
- Integration with advanced MR devices
- Smart home integration

References

- F. Vangi, D. Medeiros, M. Dastan, M. Fiorentino, MR-CoCo: An Open Mixed Reality Testbed for Co-located Couple Product Configuration and Decision-Making – A Sailboat Case Study, 2020.
- P. Milgram (1994) – Mixed Reality Taxonomy
- R. Azuma (1997) – A Survey of Augmented Reality
- M. Sereno et al. (2022) – Collaborative Work in Augmented Reality
- M. McGill et al. (2020) – Multi-user Mixed Reality Systems (VRST)
- S. Saffo et al. (2021) – Remote Collaborative VR Systems
- Epic Games (2025) – Photon Unity Networking (PUN)
- Microsoft MRTK (2023)
- Unity Netcode (2023)
- HTC VIVE Documentation (2023)

Project carried out under the guidance of
Dr. Prateek Shah & Dr. Pramit Mazumdar